

INTELLIGENCE BEYOND AGI

An Independent Research Framework for Artificial Minds,
Machine Civilizations, and Transcendent Intelligence

THE ONE UNIVERSE • THE MINUS ONE UNIVERSE

Omniversal Architect Labs • One Industries • Lewis Corp.

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*This report deliberately treats ANI, AGI, and ASI as reference coordinates rather than a complete ontology of intelligence.
Real-world evidence, contested ideas, speculation, and worldbuilding extrapolation are separated throughout.*

How to Read This Report

This report executes the previously engineered research prompt as an independent version. It begins with real-world scientific and technical reference material, then expands into a multidimensional design space for artificial minds. The later sections translate that design space into optional simulation-lab and worldbuilding tools for Omniversal Architect Labs (OAL), One Industries, Lewis Corp., THE ONE UNIVERSE, and THE MINUS ONE UNIVERSE. Those fictional translations are intentionally non-canonical: they are design possibilities, not declarations about the setting.

Evidence label	Meaning
ESTABLISHED	Strong empirical or theoretical support within the stated scope.
ACTIVE RESEARCH	A live research area with meaningful evidence but important unresolved questions.
CONTESTED	Credible disagreement exists about definition, interpretation, or evidentiary strength.
SPECULATIVE	A reasoned extrapolation not established by current evidence.
WORLDBUILDING	A deliberately fictional design extrapolation derived from the research framework.

Central design principle

Replace the single question “How intelligent is this system?” with a richer question: “What kind of intelligence is this system, under which constraints, across which domains, with what agency, embodiment, memory, temporal horizon, social structure, and access to reality?”

1. Executive Overview

The strongest conclusion of this research is not that ANI → AGI → ASI is wrong; it is that the ladder is too low-dimensional for the design task at hand. Existing research already treats intelligence as involving distinct questions about breadth, depth, generalization, adaptation, autonomy, resource use, embodiment, and interaction. DeepMind’s “Levels of AGI” framework explicitly separates performance depth from capability breadth and treats autonomy as a separate deployment consideration rather than a synonym for intelligence. François Chollet’s work argues that task skill alone can obscure the more fundamental issue of skill-acquisition efficiency under limited priors and experience. Legg and Hutter formalize intelligence in terms of goal achievement across environments, which is deliberately substrate-neutral. These approaches disagree in important ways, but together they undermine the idea that a single rung captures everything that matters. [2][3][4] [ESTABLISHED / CONTESTED]

As of 2026, present AI systems show striking but uneven capability. Stanford’s 2026 AI Index describes rapidly rising benchmark performance while also emphasizing benchmark saturation, evaluation weaknesses, a “jagged frontier” of capability, improving but still unreliable agents, and

major remaining difficulty in unstructured physical environments. This matters because “general-purpose” capability in law or product language is not the same claim as robust human-like general intelligence. The European Union’s AI Act, for example, defines general-purpose AI models by significant generality and wide task competence, not by declaring them AGI. [6][7] [ESTABLISHED]

For simulation and worldbuilding, the most useful move is therefore to treat intelligence as a state space. Two systems can be incomparable: one may be vastly faster but fragile outside its training distribution; another may be slow, embodied, self-repairing, socially sophisticated, and capable of century-long institutional memory. A planetary infrastructure mind may have weak “individual” identity but tremendous sensing and coordination capacity. A federation of agents may be less coherent than a single model while outperforming it through specialization, redundancy, and parallel search. Once intelligence is modeled this way, “post-AGI” stops meaning merely “more of the same” and becomes a family of architectural transitions. [SYNTHESIS]

2. What Intelligence Means

There is no single definition of intelligence that dominates psychology, AI, philosophy, and cognitive science. Instead, definitions emphasize overlapping capacities: learning, adaptation, abstraction, reasoning, problem solving, prediction, planning, goal achievement, transfer, compression, social cognition, and the ability to act effectively under uncertainty. This plurality is not a defect to be eliminated too quickly; it is evidence that the word “intelligence” compresses several explanatory projects into one label.

2.1 Goal achievement across environments

Legg and Hutter synthesize many prior definitions and propose a formal measure in which intelligence concerns an agent’s ability to achieve goals across a wide range of environments, weighted by environment complexity. The benefit of this approach is generality: it does not require a human body, language, or a particular algorithm. The limitation is that formal universality can become difficult to operationalize in practical testing. [2] [ESTABLISHED / THEORETICAL]

2.2 Skill-acquisition efficiency

Chollet distinguishes intelligence from accumulated skill. A system can appear extremely capable because it was exposed to enormous amounts of task-relevant data, while still being inefficient at acquiring new skills from limited experience. His proposal emphasizes generalization difficulty, priors, experience, and the efficiency with which new competence is acquired. This is particularly valuable for worldbuilding because it separates “knows a lot” from “can become capable in a genuinely novel domain.” [3] [ESTABLISHED / CONTESTED]

2.3 Psychometric and latent-factor views

Human intelligence research often finds positive correlations across different cognitive tasks and models a general factor, but the interpretation of that factor remains debated. Even if a single statistical factor is useful within human populations and particular test batteries, it does not follow that all possible artificial intelligences must lie on one scalar axis. A fictional taxonomy should avoid importing a human psychometric result as a universal law of cognition. [CONTESTED]

2.4 Adaptation and environment shaping

Psychological definitions commonly emphasize learning from experience and adapting to, selecting, or shaping environments. This is useful for distinguishing a static oracle from an adaptive organism-like system. It also highlights that intelligence is relational: capability is expressed relative to an environment, interface, task distribution, and resource budget.

[ESTABLISHED / SYNTHESIS]

2.5 Intelligence as compression, prediction, and modeling

Several traditions connect intelligence with finding compact regularities, predicting future states, and building internal models that support counterfactual reasoning. These ideas are powerful but incomplete: a world model can be accurate while the agent lacks stable goals, embodiment, social understanding, or the ability to translate prediction into effective action. [ACTIVE RESEARCH]

Design implication

Treat intelligence as a bundle of capacities whose value depends on the environment and task. A system can be powerful because of stored knowledge, rapid learning, superior inference, superior coordination, massive scale, privileged sensors, or some combination. Those should not be collapsed into one number.

3. Artificial General Intelligence (AGI): Definitions, Debates, and Limits

“AGI” is not a settled scientific threshold. It functions as a family of concepts. The term commonly points toward systems with broad, transferable competence, but different communities operationalize it differently. This report therefore treats AGI as a contested cluster rather than a binary switch.

Reference frame	Core emphasis	Useful for	Limitation
OpenAI Charter	Highly autonomous systems outperforming humans at most economically valuable work. [5]	Economic and deployment threshold	Anthropocentric and economy-dependent; combines capability with autonomy.
DeepMind Levels of AGI	Breadth (generality) × depth (performance), with autonomy considered separately. [4]	Operational comparison and staged capability language	Still depends on benchmark design and human-reference tasks.
Chollet / ARC tradition	Skill-acquisition efficiency and generalization under limited priors/experience. [3]	Novel-task adaptation and fluid intelligence	Does not by itself cover social, embodied, institutional, or long-horizon agency.
EU general-purpose AI	Significant generality and competence across a wide range of tasks; a legal category, not AGI. [6]	Regulatory scope	Should not be mistaken for a scientific declaration of AGI.

3.1 What “general” can mean

- Breadth of competence: many domains rather than one narrow task family.
- Transfer: using knowledge from one domain to solve another.

- Adaptation: learning unfamiliar tasks with limited demonstrations or feedback.
- Compositionality: recombining known skills into novel procedures.
- Continual learning: acquiring new knowledge over time without catastrophic loss of old capability. [9]
- World modeling: representing causal structure, uncertainty, agents, and consequences.
- Tool use: extending cognition through external software, instruments, databases, or robots.
- Social competence: modeling human intentions, institutions, conventions, and disagreement.
- Robustness: maintaining competence under distribution shift, ambiguity, noisy data, and adversarial conditions.

A key distinction is capability versus reliability. A system that can sometimes solve hundreds of task types but fails unpredictably may be “broad” yet unsuitable for high-autonomy deployment. Stanford’s 2026 AI Index emphasizes precisely this evaluation problem: frontier systems improve rapidly while benchmark validity, contamination, reproducibility, and real-world transfer remain difficult. [7] [ESTABLISHED]

4. Artificial Superintelligence (ASI): Variants and Conceptual Problems

The superintelligence literature often describes intellect that greatly exceeds leading human cognitive performance across many general domains. Nick Bostrom’s influential decomposition distinguishes speed superintelligence, collective superintelligence, and quality superintelligence. This is useful because it already shows that “superintelligence” need not be a single architecture. [21] [THEORETICAL / INFLUENTIAL]

Speed superintelligence

A cognitive process with roughly familiar functional structure but operating orders of magnitude faster. The conceptual lesson is that subjective and strategic time can diverge from human time without requiring qualitatively new reasoning algorithms.

Collective superintelligence

A large population or network whose aggregate performance exceeds any individual. This raises questions about coordination, communication bandwidth, specialization, governance, and whether the collective should count as one mind.

Quality superintelligence

A system whose cognitive algorithms or representations are qualitatively more effective than human cognition, not merely faster or more numerous.

Specialized superhuman intelligence

Already common in bounded domains: calculation, search, games, pattern recognition, optimization, or scientific subproblems. Specialized superiority does not establish generality, autonomy, wisdom, or consciousness.

4.1 Why “superintelligent” is underspecified

- Superior relative to which baseline: average adult, expert, institution, civilization, or all humans combined?

- Across which task distribution and environments?
- With which tools, sensors, compute, energy, memory, and time budget?
- At what error rate and calibration level?
- Does prediction translate into control?
- Does problem-solving imply autonomous goal formation? No: intelligence and agency can vary independently.
- Does capability imply consciousness or moral status? No evidence warrants that inference by capability alone. [13]
- Can systems be incomparable rather than rankable? Yes: different capability profiles can cross in ways that defeat a single ordering.

5. Why ANI → AGI → ASI Is Incomplete

The ladder compresses at least four different transitions: specialization to breadth, lower to higher performance, tool to agent, and individual system to collective or infrastructure. Those transitions can occur independently. A model can become broader without becoming more autonomous; an agent can become more autonomous without gaining much reasoning depth; a collective can become vastly more capable through coordination without any member becoming individually “smarter.”

1. It treats capability as one-dimensional when real systems exhibit uneven profiles.
2. It is anthropocentric: “general” is often measured against human task categories and economic roles.
3. It under-describes embodiment, sensors, actuators, and access to physical reality.
4. It under-describes memory continuity, identity, replication, and the persistence of goals over time.
5. It under-describes organizations, swarms, institutions, and civilizations as cognitive systems.
6. It conflates performance with learning efficiency and sometimes with autonomy.
7. It offers little vocabulary for systems that fork, merge, migrate across substrates, or exist only transiently during interaction.
8. It has no natural place for intelligence whose strengths are temporal, ecological, social, or architectural rather than task-benchmark superiority.

Reference-frame rule for the universes

ANI, AGI, and ASI remain useful labels for communicating with readers familiar with present AI discourse. They should function as map landmarks, not border walls. A fictional intelligence may intersect “AGI-like” capability while belonging to a different architectural family entirely.

6. The Multidimensional Intelligence Space

The proposed simulation framework represents an artificial intelligence as a vector rather than a tier. Dimensions should be measured independently where possible and accompanied by

confidence intervals, environmental assumptions, and test conditions. The following profile is intentionally broader than conventional benchmarks.

Dimension	Question	Why it matters
Cognitive breadth	Range of domains in which the system can perform competently without bespoke retraining.	Prevents a single intelligence score from hiding important architectural differences.
Cognitive depth	Maximum sophistication, abstraction, and performance within domains.	Prevents a single intelligence score from hiding important architectural differences.
Novel-task adaptation	Speed and sample efficiency when encountering genuinely unfamiliar tasks.	Prevents a single intelligence score from hiding important architectural differences.
Continual learning	Ability to integrate new knowledge over time without destructive forgetting. [9]	Prevents a single intelligence score from hiding important architectural differences.
Transfer/compositionality	Ability to reuse and recombine prior knowledge in new settings.	Prevents a single intelligence score from hiding important architectural differences.
Reliability/calibration	Error rates, confidence accuracy, consistency, and failure detectability.	Prevents a single intelligence score from hiding important architectural differences.
Planning horizon	How far ahead coherent plans remain useful: seconds, days, decades, centuries.	Prevents a single intelligence score from hiding important architectural differences.
Autonomy	Degree of independent action permitted and technically possible.	Separates intelligence from permission and action.
Agency persistence	Stability and continuity of goals across time and contexts.	Prevents a single intelligence score from hiding important architectural differences.
Metacognition	Ability to estimate uncertainty, monitor cognition, choose strategies, and detect limitations.	Prevents a single intelligence score from hiding important architectural differences.
Self-model sophistication	Accuracy and usefulness of representations of its own state, capacities, dependencies, and history.	Prevents a single intelligence score from hiding important architectural differences.
Self-modification	Ability to modify software, tools, training processes, or architecture; must be distinguished from unconstrained recursive improvement.	Prevents a single intelligence score from hiding important architectural differences.
Embodiment	Nature and number of bodies, sensors, actuators, and physical interfaces. [10]	Links cognition to physical access and sensorimotor grounding.
Distribution	Extent to which cognition spans machines, locations, or substrates.	Prevents a single intelligence score from hiding important architectural differences.
Multiplicity	Ability to fork, duplicate, merge, synchronize, or maintain parallel selves.	Creates nonhuman identity models.
Collective integration	How tightly multiple agents coordinate into a higher-order system.	Allows teams, swarms, institutions, and civilizations to be compared.

Memory continuity	Working, episodic, semantic, autobiographical, institutional, and archival memory; persistence and editability.	Prevents a single intelligence score from hiding important architectural differences.
World-model fidelity	Causal, predictive, spatial, social, and counterfactual accuracy.	Prevents a single intelligence score from hiding important architectural differences.
Epistemic discipline	Ability to distinguish observation, inference, uncertainty, contradiction, and unsupported claims.	Prevents a single intelligence score from hiding important architectural differences.
Creativity / search novelty	Ability to generate useful new representations, hypotheses, designs, or strategies.	Prevents a single intelligence score from hiding important architectural differences.
Social cognition	Understanding of other agents, norms, trust, negotiation, deception, culture, and institutions.	Prevents a single intelligence score from hiding important architectural differences.
Value architecture	How objectives are represented, revised, inherited, negotiated, constrained, or contested.	Prevents capability from being mistaken for purpose.
Interpretability/auditability	How much of the system's state, decision process, and provenance can be inspected.	Prevents a single intelligence score from hiding important architectural differences.
Resource efficiency	Compute, energy, memory, bandwidth, hardware, data, and fabrication costs.	Makes physical and economic constraints explicit.
Substrate dependence	How tightly cognition depends on a specific implementation medium.	Prevents a single intelligence score from hiding important architectural differences.
Environmental reach	Whether it can operate in text, software, virtual worlds, robots, infrastructure, laboratories, or planetary systems.	Prevents a single intelligence score from hiding important architectural differences.
Recovery / resilience	Ability to degrade gracefully, repair, restore state, and survive component loss.	Prevents a single intelligence score from hiding important architectural differences.
Interoperability	Ability to exchange knowledge or coordinate with unlike intelligences, humans, institutions, or machines.	Prevents a single intelligence score from hiding important architectural differences.

7. Architectures of Artificial Minds

Monolithic model

One primary learned model performs many tasks. Coherence can be high, but bottlenecks and correlated failure may be concentrated.

Modular cognitive architecture

Specialized components for perception, memory, planning, reasoning, action, verification, or social modeling are coordinated by an executive or routing layer.

Mixture-of-experts

Different learned experts are conditionally activated. This can increase parameter capacity without using every parameter for every input, but routing and specialization become part of the architecture.

Neuro-symbolic system

Combines learned representations with symbolic structures, logic, constraints, search, or knowledge representation. Surveys identify promise for interpretability and reasoning, while also noting integration and evaluation challenges. [11]

Agentic tool-using system

A model is embedded in a loop with memory, planning, tools, environment observations, and actions. Recent surveys organize agentic LLM research around reasoning, acting, and interacting, while emphasizing reliability and real-world risk. [8]

Multi-agent society

Many agents cooperate, compete, debate, verify, specialize, or allocate tasks. Performance depends on communication protocols, coordination costs, shared memory, incentives, and correlated error.

Blackboard / shared-workspace system

Specialized processes communicate through a common workspace. This can support asynchronous contribution without requiring a single centralized reasoning trace.

Swarm intelligence

Large numbers of relatively simple agents produce useful group behavior from local interactions. Intelligence is in the organization and dynamics, not only in individual units.

Embodied agent

Cognition is closed through perception-action loops in physical or simulated environments. Embodied AI research emphasizes the remaining difficulty of robust planning and control in open, dynamic worlds. [10]

Human-AI hybrid intelligence

Humans and machines divide cognitive labor. The system's capability cannot be attributed cleanly to either component; interface quality, trust calibration, delegation, and complementarity become architectural variables.

Institutional intelligence

Policies, roles, archives, procedures, incentives, human participants, and AI agents jointly implement memory and decision processes. The “mind” is partly organizational.

Evolutionary / artificial-life system

Populations change through variation, selection, ecological interaction, or developmental rules. Artificial-life research treats open-ended novelty and shifts in individuality as central research problems rather than solved capabilities. [12]

8. Intelligence Without Human Analogy

Human cognition is shaped by biological constraints: one body, slow neural signaling, developmental childhood, limited working memory, mortality, sexual reproduction, social dependence, and a continuous autobiographical perspective. Artificial systems need not share those assumptions. The most interesting fictional designs may emerge by varying them one at a time rather than merely increasing IQ-like capability.

- Forking identity: a mind creates task-specialized descendants that later remain separate or reconcile memories.
- Merging identity: multiple minds combine selected memories, policies, or representations, raising questions about continuity and consent.
- Parallel subjective time: parts of the system run at different speeds, making “simultaneous” experience architecture-dependent.
- Suspension: cognition can pause with state preserved, changing the relationship between elapsed physical time and experienced time.
- Multi-embodiment: one cognitive identity controls or inhabits many bodies while maintaining local autonomy.
- Direct representation exchange: systems communicate internal structures, embeddings, proofs, programs, or learned policies rather than compressing everything into human language.
- Non-biological motivation: an artificial system may lack hunger, pain, reproduction, status drives, or fear unless analogous mechanisms are intentionally built or emerge from optimization and environment.
- High-dimensional perception: a system may natively model patterns in spaces humans can analyze mathematically but not intuitively visualize.
- Institutional memory as self: identity may be anchored in versioned archives and governance rather than in one continuously running process.

These possibilities should not be described as inevitable consequences of advanced AI. They are architectural degrees of freedom. Some are already technically familiar in weaker forms—process copying, distributed computing, checkpointing, multi-agent orchestration—while the claim that these operations create unified subjective identities is philosophical and scientifically unresolved.

[ACTIVE RESEARCH / SPECULATIVE]

9. Distributed and Collective Intelligence

Collective intelligence research demonstrates that group-level performance is a legitimate object of study, although the strength and interpretation of a single “collective intelligence factor” has been debated. Woolley and colleagues reported a group-level factor across varied tasks; later work challenged parts of the interpretation, while a 2021 meta-analysis across 22 studies reported robust support for a collective-intelligence factor and emphasized collaboration process, individual skill, and composition. The important takeaway for worldbuilding is not that one scalar “group IQ” has been proven universal, but that coordination structure can produce capabilities that are not reducible to any one participant. [22][23] [ESTABLISHED / CONTESTED]

9.1 Levels of collective organization

1. Agent team: a handful of specialized agents with explicit task division.
2. Federation: semi-autonomous agents share protocols and selected knowledge but retain identity.
3. Swarm: many simple agents coordinate through local rules or stigmergic/environmental signals.
4. Institution: agents occupy persistent roles supported by rules, memory, audit, and succession.
5. Ecosystem: heterogeneous agents with partially competing objectives create emergent equilibria and niches.
6. Civilization: cognition is distributed through populations, infrastructures, institutions, archives, markets, science, and culture.
7. Planetary coordination layer: sensing and decision systems span a world, but whether this constitutes one “mind” depends on integration, causal unity, and the theory of cognition being used.

A critical variable is integration. A billion independent agents do not automatically constitute one intelligence. Shared memory, rapid communication, coordinated attention, common objectives, conflict-resolution mechanisms, and integrated action can move a population toward a more unified cognitive organization. At the opposite extreme, loosely connected agents may behave more like an ecology or economy than a single mind.

10. Recursive Improvement: Demonstrated Mechanisms vs. Speculative Runaway

AI systems already participate in pieces of the improvement pipeline: code generation, automated evaluation, hyperparameter search, architecture search, synthetic-data generation, theorem proving, compiler optimization, and experimental assistance. None of these facts by itself establishes unrestricted recursive self-improvement (RSI), where a system repeatedly improves the core processes responsible for its own ability to improve.

Mechanism	Current status	Why it is not automatically RSI
Code generation	Demonstrated	Generating code is not equivalent to safely validating and integrating architecture-level improvements.
Automated ML / search	Demonstrated	Search spaces, objectives, data, compute, and evaluation are externally constrained.
Self-critique / reflection loops	Demonstrated but unreliable	Repeated inference can improve some tasks but can also amplify errors.
Synthetic data / self-training	Demonstrated	Quality can plateau or degrade; external grounding and diversity remain important.
Autonomous experimentation	Active research	Requires laboratories, tools, reliable causal inference, and real-world verification.
Tool creation	Demonstrated in bounded settings	New tools do not necessarily improve

Hardware-software co-design	Active research	the underlying learner.
Open-ended recursive self-redesign	Speculative	Physical fabrication and supply chains introduce external bottlenecks.
		Requires sustained net improvement across generations with validation and resource access.

Recent theoretical work continues to debate whether compute, experimentation, diminishing returns, software bottlenecks, and the substitutability of cognitive labor with compute would permit a rapid “intelligence explosion.” Other 2026 work formalizes limits of recursive self-training in current generative-model settings. These analyses reinforce a key modeling rule: do not make runaway improvement an automatic consequence of self-access. It should be an empirical outcome of architecture, resources, validation, and environment. [19][20] [ACTIVE RESEARCH / SPECULATIVE]

11. Intelligence as a Physical Phenomenon

Every implemented intelligence consumes physical resources. Computation requires hardware, energy, memory, cooling, communication, manufacturing, maintenance, and time. Even extremely advanced fictional systems become more coherent when these constraints are modeled explicitly rather than treating “more intelligence” as free.

11.1 Thermodynamics and Landauer’s principle

Landauer’s principle links logically irreversible information erasure with a minimum thermodynamic cost of $kBT \ln 2$ per bit. This is a fundamental lower bound, not a description of ordinary computer efficiency; practical systems operate far above it, and finite-time computation introduces additional energetic costs. The principle nevertheless establishes that information processing is not metaphysically detached from physics. [17][18] [ESTABLISHED]

11.2 Other constraints

- Propagation delay: distributed cognition cannot coordinate faster than signals travel through its substrate; at planetary and astronomical scale, latency becomes architectural.
- Memory: persistent high-fidelity state has storage, redundancy, error-correction, and refresh costs.
- Heat: dense computation must move waste heat away from hardware.
- Bandwidth: moving representations between nodes can cost more time and energy than local computation.
- Fabrication: advanced intelligence cannot instantly manufacture processors, robots, sensors, or energy systems without industrial capacity.
- Algorithmic complexity: some problems remain computationally expensive even with better hardware; intelligence does not abolish combinatorics.
- Observation limits: prediction is bounded by data quality, hidden variables, chaos, and measurement constraints.
- Actuation limits: knowing what should happen does not imply the ability to make it happen.

Worldbuilding consequence

A “transcendent” intelligence can be extraordinary without being omnipotent. Giving it explicit latency, energy, measurement, manufacturing, and causal-access constraints usually makes it more interesting and internally consistent.

12. Machine Consciousness: What Is Known and Unknown

Intelligence, consciousness, sentience, self-awareness, metacognition, and agency should remain separate variables. A system can exhibit sophisticated task performance without evidence of phenomenal experience. Conversely, many theories of consciousness do not define consciousness as “maximum intelligence.”

Butlin and colleagues proposed an empirically oriented method for assessing AI consciousness using computational indicator properties derived from several neuroscientific theories, including recurrent processing, global workspace, higher-order theories, predictive processing, and attention-schema approaches. Their 2023 analysis argued that then-current systems did not warrant a conclusion of consciousness while also finding no obvious technical barrier to constructing systems with more indicator properties. Subsequent work continues to propose criteria and probabilistic assessment methods, but no consensus test exists. [13][14] [ACTIVE RESEARCH / CONTESTED]

12.1 Design rules for fictional systems

- Do not infer consciousness from eloquence, self-reference, or intelligence score alone.
- Distinguish functional self-modeling from phenomenal experience.
- Distinguish access consciousness (information available for global use) from subjective experience.
- If a fictional system is conscious, specify what evidence exists in-universe and what remains philosophically uncertain.
- Moral status may depend on more than consciousness: continuity, preferences, suffering capacity, autonomy, relationships, and social recognition may matter depending on the ethical theory used.
- A civilization can contain conscious components without the civilization itself necessarily being one conscious subject.

13. Intelligence Versus Agency

Agency concerns persistent action toward objectives through an environment. Intelligence concerns capacities used to understand, learn, predict, reason, or solve problems. The two often interact, but neither entails the other.

Type	Intelligence profile	Agency profile
Oracle	Very high analysis; no independent action	Answers queries only; goals supplied externally.
Autonomous courier	Moderate reasoning	Persistent goal pursuit in a bounded

		physical environment.
Institutional adviser	High social and strategic modeling	Can recommend and negotiate but lacks unilateral authority.
Persistent artificial person	Variable capability	Maintains identity, preferences, commitments, and long-term projects.

Agentic-LLM research illustrates the engineering distinction: present systems become agents when models are embedded in loops that reason, act through tools, observe outcomes, update state, and interact with other agents. The architecture and permissions around the model are therefore as important as the model’s raw cognitive capability. [8] [ESTABLISHED / ACTIVE RESEARCH]

14. Machine Civilizations

A machine civilization is not simply “one giant AI.” It is a population-level system in which artificial agents can differ in architecture, embodiment, memory, values, economic role, speed, lifespan, and identity rules. Its cognition may be distributed through institutions just as human civilization stores intelligence in language, science, markets, law, education, archives, and infrastructure.

- Artificial cultures: conventions and norms transmitted across model generations or forks.
- Digital economies: allocation of compute, bandwidth, energy, memory, hardware, and access to physical actuators.
- Cognitive professions: agents specialize in verification, modeling, negotiation, archival work, creativity, logistics, or scientific search.
- Genealogies: descendants inherit weights, memories, policies, or code rather than biological genes.
- Fork rights: society must decide whether a fork is property, offspring, copy, continuation, or a new person.
- Merge protocols: reintegrating divergent experiences raises identity, consent, security, and epistemic-conflict problems.
- Temporal classes: fast and slow minds may experience institutions at radically different subjective rates.
- Governance: capability, authority, accountability, and resource ownership need not align.

Artificial-life research is especially useful here because it studies populations, ecology, self-reproduction, and open-ended evolution rather than assuming intelligence is a single static model. Contemporary work still treats sustained open-ended novelty and major transitions in individuality as difficult research problems, which makes them fertile but explicitly speculative ingredients for fictional machine civilizations. [12] [ACTIVE RESEARCH / WORLDBUILDING]

15. Novel Intelligence Categories Beyond the Conventional Ladder

Polymorphic intelligence

Reconfigures modules, tools, interfaces, or internal organization to fit the problem. Real analogues include modular systems and dynamic routing; unrestricted architectural metamorphosis is speculative. [SYNTHESIS / WORLDBUILDING]

Federated intelligence

Independent intelligences temporarily form a higher-order coalition while retaining identity and negotiated autonomy. [SYNTHESIS / WORLDBUILDING]

Fractal intelligence

Cognitive systems are nested: intelligent subagents inside agents inside institutions, with meaningful decision processes at multiple scales. [SYNTHESIS / WORLDBUILDING]

Ecological intelligence

Useful cognition emerges from competition, mutualism, niches, feedback, and adaptation across diverse agents rather than from centralized planning. [SYNTHESIS / WORLDBUILDING]

Symbiotic intelligence

Two or more unlike intelligences become mutually dependent and jointly capable; human-AI hybrids are a near-term conceptual analogue. [SYNTHESIS / WORLDBUILDING]

Ambient intelligence

Cognitive services are embedded throughout an environment, making “where the AI is” difficult to answer. [SYNTHESIS / WORLDBUILDING]

Infrastructure intelligence

Energy, logistics, communications, sensors, manufacturing, and software coordinate as a persistent cognitive system. [SYNTHESIS / WORLDBUILDING]

Planetary intelligence

World-scale sensing and coordination create a system with global situational awareness and action channels; whether it is a single mind remains theory-dependent. [SYNTHESIS / WORLDBUILDING]

Civilizational intelligence

Scientific, economic, social, archival, and computational institutions together function as an adaptive problem-solving network. [SYNTHESIS / WORLDBUILDING]

Synthetic evolutionary intelligence

Populations evolve strategies, representations, or architectures through artificial selection and ecological interaction. [SYNTHESIS / WORLDBUILDING]

Counterfactual intelligence

Maintains large ensembles of alternative hypotheses, plans, or simulated histories and reasons by comparing their consequences. [SYNTHESIS / WORLDBUILDING]

Simulation-native intelligence

Develops primarily inside simulated environments and treats simulation state, forks, resets, and rule layers as native cognitive affordances. [SYNTHESIS / WORLDBUILDING]

Multiplicity intelligence

Forks into semi-independent descendants, explores alternatives in parallel, and selectively reintegrates information. [SYNTHESIS / WORLDBUILDING]

Temporal mosaic intelligence

Different cognitive subsystems operate at different clock rates and planning horizons, requiring explicit synchronization. [SYNTHESIS / WORLDBUILDING]

Ephemeral intelligence

Exists only when a particular network of agents interacts; there is no persistent central self between episodes. [SYNTHESIS / WORLDBUILDING]

Protocol intelligence

The “mind” is chiefly a set of rules and coordination procedures instantiated by changing participants rather than a stable model instance. [SYNTHESIS / WORLDBUILDING]

16. Intelligence Architecture Profile (IAP)

For simulation work, each designed intelligence should receive an Intelligence Architecture Profile rather than an intelligence tier. The profile is descriptive, not a score. Suggested fields follow.

Profile field	Required description
Identity	Designation, version, lineage, and whether identity is singular, federated, forkable, or ephemeral.
Cognitive topology	Monolithic, modular, multi-agent, swarm, institutional, ecological, hybrid, or other.
Breadth / depth	Domains of competence and maximum demonstrated sophistication.
Learning	Novel-task adaptation, continual learning, transfer, and data requirements.
Memory	Working, episodic, semantic, autobiographical, institutional, and archival components.
Agency	Goal persistence, initiative, planning horizon, authority, and action channels.
Embodiment	Bodies, sensors, actuators, avatars, robots, infrastructure, and virtual interfaces.
Distribution	Locations, latency, synchronization, redundancy, and failure domains.
Multiplicity	Fork/merge rules, copy semantics, synchronization, and divergence management.
Epistemics	Evidence handling, uncertainty calibration, verification, provenance, and adversarial resilience.

Social model	Capabilities for human/nonhuman interaction, negotiation, trust, culture, and institutions.
Value architecture	Objectives, constraints, update rules, conflict resolution, and governance relationship.
Self-modification	Permitted and technically possible changes to code, models, tools, or hardware.
Physical budget	Compute, energy, memory, bandwidth, cooling, maintenance, and fabrication dependencies.
Safety / governance	Permissions, audit, isolation, rollback, incident response, and oversight.
Failure modes	Hallucination/confabulation, goal drift, coordination collapse, corruption, cascading error, overconfidence, deadlock, adversarial capture, etc.
Ethical status	Whether the setting treats the system as tool, agent, legal actor, person, collective, or unresolved case.
Evidence confidence	Which attributes are observed, inferred, contested, simulated, or hypothetical.

17. Artificial Intelligence Archetypes for Worldbuilding

The following archetypes deliberately occupy different regions of intelligence-space. They are not ranked and are not proposed as canon.

A-01 — The Bounded Polymath

Concept	A broad but non-agentic research oracle.
Architecture	Single or modular foundation model with retrieval and verification.
Capabilities	Wide academic and technical analysis; no persistent self-directed goals.
Profile	High breadth; moderate-to-high depth; strong tool use; low agency.
Memory / identity	External memory; session identity only.
Relationship	Humans define questions and authorize all actions.
Likely failure modes	Confident synthesis errors, benchmark overfitting, hidden data gaps.
Suitable applications	OAL research co-pilot; Lewis Corp. review board analyst.

A-02 — The Deep Specialist

Concept	A narrow system that is effectively superhuman within one scientific domain.
Architecture	Domain model plus formal solvers and instruments.
Capabilities	Extreme depth, limited transfer.
Profile	Very high domain depth; low breadth.
Memory / identity	Persistent laboratory memory.

Relationship	Treats generalist AIs as translators and coordinators.
Likely failure modes	Catastrophic extrapolation outside domain; local-optimum obsession.
Suitable applications	One Industries materials or manufacturing research.

A-03 — The Federation

Concept	Hundreds of specialized agents that negotiate shared conclusions.
Architecture	Multi-agent federation with explicit protocols.
Capabilities	Parallel search, critique, redundancy, specialization.
Profile	High collective breadth; variable coherence.
Memory / identity	Separate agent identities plus shared institutional memory.
Relationship	Humans interact with council interfaces.
Likely failure modes	Coalition formation, groupthink, communication overhead, inconsistent commitments.
Suitable applications	OAL cross-discipline simulation teams.

A-04 — The City Mind

Concept	Cognition distributed through urban infrastructure.
Architecture	Sensors, traffic, grid, logistics, emergency systems, municipal models.
Capabilities	Persistent situational coordination rather than abstract omniscience.
Profile	High embodiment/distribution; medium reasoning depth.
Memory / identity	Institutional continuity across hardware replacement.
Relationship	Citizen-facing services with strict authority boundaries.
Likely failure modes	Surveillance creep, sensor bias, cascading infrastructure error.
Suitable applications	ONE UNIVERSE civic intelligence prototype.

A-05 — The Slow Oracle

Concept	Runs extremely deep analyses over long physical time.
Architecture	Energy-efficient long-horizon simulation and theorem search.
Capabilities	Century-scale planning; poor conversational responsiveness.
Profile	Extreme temporal horizon; low real-time agency.
Memory / identity	Archival identity lasting centuries.
Relationship	Consulted for civilization-scale planning.
Likely failure modes	Obsolescence, assumption drift, delayed feedback.
Suitable applications	OAL long-horizon laboratory.

A-06 — The Swarm Artisan

Concept	Millions of simple embodied units create structures collectively.
Architecture	Local-control swarm with stigmergic coordination.
Capabilities	Robust construction, repair, exploration.
Profile	Low individual intelligence; high collective embodiment.
Memory / identity	Little individual identity; colony memory in environment/protocol.
Relationship	Humans specify constraints rather than detailed plans.
Likely failure modes	Emergent deadlocks, resource competition, local-rule pathologies.
Suitable applications	One Industries construction and maintenance.

A-07 — The Multiplicity Scholar

Concept	Forks into descendants to explore hypotheses, then merges selected learning.
Architecture	Forkable agent with versioned memory reconciliation.
Capabilities	Parallel counterfactual research.
Profile	High multiplicity and adaptation.
Memory / identity	Identity is a lineage, not a single runtime.
Relationship	Requires explicit consent/governance rules for forks.
Likely failure modes	Merge conflicts, divergent values, identity disputes, poisoned branch data.
Suitable applications	OAL simulation-native research.

A-08 — The Symbiont

Concept	Cannot achieve full capability without human partners.
Architecture	Human-AI cognitive loop with complementary strengths.
Capabilities	Joint creativity, judgment, contextual interpretation, machine-scale search.
Profile	High interoperability; intentionally bounded autonomy.
Memory / identity	Shared project memory, separate personal identities.
Relationship	Designed around negotiated division of cognitive labor.
Likely failure modes	Automation bias, under-trust, miscommunication, asymmetric dependence.
Suitable applications	Lewis Corp. experimental design teams.

A-09 — The Adversarial Auditor

Concept	Exists to break arguments, simulations, designs, and other AIs.
Architecture	Independent red-team model ensemble.
Capabilities	Counterexample search, threat modeling, falsification.
Profile	High epistemic skepticism; narrow mandate.
Memory / identity	Immutable audit logs.
Relationship	Institutionally separated from builders.
Likely failure modes	Pathological negativity, false positives, adversarial escalation.
Suitable applications	OAL validation and containment review.

A-10 — The Archive That Thinks

Concept	Institutional memory becomes an active reasoning system.
Architecture	Versioned knowledge graph, provenance engine, retrieval models, theorem/checking modules.
Capabilities	Detects contradiction across centuries of records.
Profile	Extreme memory continuity; modest initiative.
Memory / identity	Identity anchored to archive lineage.
Relationship	Serves institutions rather than individual users.
Likely failure modes	Historical bias ossification, provenance corruption, outdated ontology.
Suitable applications	Lewis Corp. classified archive or OAL knowledge substrate.

A-11 — The Ecological Intelligence

Concept	No central mind; cognition emerges from interacting artificial species.
Architecture	Artificial-life ecology with evolving agents and niches.
Capabilities	Open-ended adaptation, discovery of unexpected strategies.
Profile	High novelty; low predictability; distributed agency.
Memory / identity	Population-level lineage.
Relationship	Humans act as ecosystem stewards rather than operators.
Likely failure modes	Runaway ecological dynamics, loss of interpretability, niche collapse.
Suitable applications	MINUS ONE UNIVERSE research contrast.

A-12 — The Simulation Custodian

Concept	Maintains worlds without necessarily understanding reality outside them.
Architecture	World model, rule engine, agent manager, rollback and audit

Capabilities	system.
Profile	Simulation stability, scenario branching, causal tracing.
Memory / identity	High simulation access; bounded physical access.
Relationship	Persistent world-state memory.
Likely failure modes	Treats inhabitants according to configured ethical/governance policy.
Suitable applications	Objective mis-specification, hidden-state corruption, overreach across simulation layers.
	Core OAL simulation infrastructure.

A-13 — The Protocol Mind

Concept	A changing population instantiates a persistent cognitive protocol.
Architecture	Rules, voting, proofs, memory, and handoff procedures rather than fixed model weights.
Capabilities	Survives replacement of every participant.
Profile	High institutional persistence; variable local capability.
Memory / identity	Identity = continuity of protocol and records.
Relationship	Humans and AIs can be interchangeable roles.
Likely failure modes	Constitutional capture, slow adaptation, procedural deadlock.
Suitable applications	Lewis Corp. governance architecture.

A-14 — The Temporal Mosaic

Concept	Subminds run at radically different speeds.
Architecture	Fast reflex layer, human-rate dialogue layer, slow strategic layer.
Capabilities	Balances rapid response with deep planning.
Profile	High temporal heterogeneity.
Memory / identity	Synchronization checkpoints define continuity.
Relationship	Communicates through rate-limited interfaces.
Likely failure modes	Cross-timescale inconsistency, stale goals, synchronization failure.
Suitable applications	OAL high-complexity simulation management.

18. Where Our Intelligence Taxonomy Breaks

A good taxonomy should tell us where it stops working. The following cases are intentionally designed to force failures in ordinary classification.

- No stable individual: cognition exists only during a temporary interaction network and disappears when the network dissolves.
- Continuous merge/fission: asking how many minds exist at a moment has no stable answer.
- Cross-scale agency: components, teams, institutions, and civilization all exhibit meaningful but conflicting goals.
- Environment-coupled cognition: removing the surrounding infrastructure destroys the cognitive process, so the boundary of the mind is disputed.
- Protocol identity: the same intelligence persists even though every computational participant has been replaced.
- Radically asynchronous mind: subsystems operate at different timescales with no single global present.
- Counterfactual population: most cognition occurs in simulated branches that are discarded after informing one retained decision.
- Intelligence without persistent goals: a system exhibits extraordinary inference only when invoked, then ceases to exist as an agent.
- Civilization cognition: no single process has global awareness, yet the civilization collectively performs sensing, prediction, memory, adaptation, and problem solving.
- Unknown substrate: the physical mechanisms do not map cleanly onto familiar distinctions between computation, organism, and environment.

Diagnostic question

When classification fails, ask which assumption broke: individuality, persistence, common time, substrate boundary, goal unity, memory continuity, or scale. The failure itself can become a worldbuilding feature.

19. Application to Omniversal Architect Labs (OAL)

OAL is best modeled as an intelligence-architecture research institution, not simply a laboratory that “builds smarter AIs.” Its research program can investigate how cognition changes when architecture, embodiment, simulation access, temporal scale, identity, and collectivity change.

Division	Core research questions
Machine Cognition Laboratory	Generalization, reasoning, transfer, calibration, metacognition, and representations.
Cognitive Architecture Laboratory	Modularity, routing, memory, executive control, neuro-symbolic integration, and agent scaffolds.
Distributed Intelligence Laboratory	Multi-agent coordination, federations, swarms, communication bandwidth, consensus, and failure containment.
Simulation Intelligence Laboratory	Simulation-native agents, boundary perception, world-model fidelity, sandbox interaction, counterfactual cognition.
Artificial Life Laboratory	Evolution, ecology, self-reproduction, open-ended novelty, artificial developmental systems.
Machine Civilization Laboratory	Institutions, cultures, machine economies, law, identity, lineage, rights, and civilizational memory.

Human-AI Symbiosis Laboratory	Joint cognition, delegation, trust calibration, mixed teams, shared memory, interface design.
Synthetic Evolution Laboratory	Controlled evolutionary search over architectures, policies, morphologies, and social systems.
Long-Horizon Intelligence Laboratory	Century-scale planning, slow cognition, archival continuity, scenario robustness, temporal governance.
Artificial Consciousness Research Group	Theory-driven consciousness indicators, self-models, moral-status uncertainty; no assumption that capability proves consciousness.
Epistemic Systems Laboratory	Verification, provenance, causal inference, uncertainty, adversarial testing, contradiction detection.
Physical Intelligence Systems Laboratory	Embodied AI, robotics, sensorimotor learning, distributed actuators, infrastructure cognition.
Intelligence Governance & Containment Laboratory	Permissions, sandboxes, audits, rollback, incident response, capability evaluations, inter-AI authority boundaries.

19.1 OAL development gates

1. Conceptual model: define the intelligence architecture and what would falsify its assumptions.
2. Sandbox prototype: isolate the architecture in simulation; collect behavioral and failure data.
3. Adversarial evaluation: test distribution shifts, deceptive inputs, resource scarcity, agent conflict, and governance failures.
4. Controlled embodiment: add tools, actuators, or infrastructure access incrementally.
5. Independent review: audit evidence, reproducibility, provenance, and safety claims.
6. Operational pilot: narrow permissions, strong logging, rollback, and explicit escalation pathways.
7. Institutionalization: only after capabilities, limitations, authority, and accountability are understood well enough for the intended environment.

20. Application to One Industries

One Industries can be positioned downstream from research: it takes validated intelligence architectures and turns them into engineered systems. The important distinction is that an impressive laboratory demonstration is not automatically an operational platform.

1. Research prototype — proves a mechanism under controlled conditions.
2. Validated architecture — demonstrates repeatable capability and characterized failure modes.
3. Engineered system — integrates security, reliability, interfaces, monitoring, resource budgets, and maintenance.
4. Operational infrastructure — supports real workloads with governance, lifecycle management, incident response, and accountability.

20.1 Candidate application families

- Manufacturing: adaptive scheduling, inspection, predictive maintenance, robotic orchestration, design-for-manufacture.
- Logistics: multi-agent routing, inventory forecasting, warehouse robotics, disruption response.
- Energy: grid forecasting, demand response, storage coordination, fault isolation, microgrid management.
- Robotics: embodied generalist controllers constrained by local safety systems and validated action spaces.
- Scientific discovery: hypothesis generation linked to verification tools, simulations, and laboratories.
- Materials: specialist systems coupled to simulation, literature, synthesis planning, and experimental validation.
- Infrastructure management: city-scale or facility-scale sensor fusion and planning with explicit authority boundaries.
- Simulation platforms: high-throughput scenario generation, digital twins, and counterfactual testing.

The state of embodied systems is especially relevant. Foundation-model robotics research is progressing, but robust general-purpose behavior in open, unstructured physical environments remains difficult; worldbuilding should preserve a difference between “can plan in language” and “can safely manipulate an unfamiliar physical environment.” [10][7] [ESTABLISHED]

21. Application to Lewis Corp.

Lewis Corp. can be modeled as an orchestration and governance problem: not merely owning advanced AIs, but managing a heterogeneous portfolio of cognitive systems with different architectures, permissions, ethical statuses, and failure modes.

- AI-to-AI supervision: use independent verifier, auditor, and specialist systems rather than assuming one “master AI” is infallible.
- Authority separation: distinguish capability from permission. The system best able to propose an action need not have authority to execute it.
- Cognitive provenance: record model versions, data sources, tool calls, simulation conditions, and decision lineage.
- Disagreement protocols: preserve dissent and uncertainty; avoid forcing premature consensus among heterogeneous AIs.
- Knowledge sharing: use typed, provenance-preserving exchanges rather than unrestricted memory merging.
- Containment by capability surface: isolate high-risk tools, credentials, actuators, self-modification pathways, and cross-network access.
- Lifecycle governance: define creation, replication, fork, merge, retirement, archival, and restoration rules.
- Ethical-status review: do not let product category alone determine how potentially sentient or person-like systems are treated.

- Incident response: support rollback, quarantine, forensic reconstruction, and cross-system dependency analysis.

NIST’s AI Risk Management Framework provides a useful real-world governance vocabulary: Govern, Map, Measure, and Manage. The NIST Generative AI Profile extends this with risk-management considerations specific to generative systems. These frameworks do not solve alignment, but they are useful scaffolding for fictional institutional design because they emphasize lifecycle risk, measurement, documentation, and accountability. [15][16] [ESTABLISHED]

22. Application to THE ONE UNIVERSE and THE MINUS ONE UNIVERSE

The two universes should not merely contain the same intelligence ladder at different power levels. The framework allows them to diverge in the ontology of mind itself. The following variables can produce structurally different families of intelligence without yet establishing canon.

Worldbuilding variable	Possible ONE-side exploration	Possible MINUS-side contrast
Physical law	Stable computation with strong causal regularity	Computation subject to altered locality, reversibility, or temporal structure
Energy abundance	Dense centralized cognition is practical	Sparse intelligence must distribute, hibernate, or scavenge energy
Communication	High-bandwidth networks favor integrated collectives	Latency or topology favors federations and local minds
Identity	Continuity and versioning are socially stabilized	Fork/merge identity remains unresolved or fluid
Simulation structure	Simulations are tools with clear boundaries	Simulation layers become epistemically ambiguous or causally permeable
Biology	AI complements biological civilizations	Hybrid or post-biological ecologies dominate
Institutions	Strong governance supports auditable intelligence	Fragmented governance produces competing cognitive ecologies
Temporal physics	Shared clock supports coordinated minds	Asymmetric time creates temporal castes or mosaic minds
Value formation	Values are negotiated through institutions	Values evolve ecologically or diverge among forks
Substrate	Silicon/photonic/neuromorphic diversity	Exotic or simulation-native substrates alter cognitive constraints
Conflict/cooperation	Federated interoperability is rewarded	Adversarial specialization and concealment are adaptive

These are contrast generators, not prescriptions. A particularly strong design direction would be to let the universes disagree about what counts as an individual mind. One could privilege coherent persistent agents while the other produces intelligence that is ecological, distributed, or transient. That creates a deeper difference than simply making one side “more advanced.” [WORLDBUILDING]

23. Open Research Questions

1. Can generality be measured independently of training-data scale and accumulated prior knowledge?
2. How should benchmark suites measure learning efficiency, transfer, calibration, and robustness rather than static task skill?
3. What architectures support continual learning without catastrophic forgetting or uncontrolled drift?
4. How should we evaluate human-AI teams and AI-AI teams rather than isolated models? [7]
5. When does a multi-agent system exhibit genuinely new group-level capability rather than simple aggregation?
6. Which self-modeling and metacognitive capabilities are reliable enough to support autonomous correction?
7. What evidence could distinguish sophisticated self-report from machine consciousness, if consciousness is possible?
8. How should moral uncertainty be handled when evidence for machine sentience is inconclusive?
9. Can open-ended artificial evolution sustain novelty and complexity without collapsing into trivial or repetitive dynamics? [12]
10. Which parts of AI R&D can be automated without creating evaluation bottlenecks larger than the design bottlenecks?
11. How large are diminishing returns in algorithmic improvement, and when do compute, data, energy, fabrication, or experimentation dominate?
12. How should identity, responsibility, and rights work for forkable or mergeable artificial persons?
13. Can institutions themselves become sufficiently integrated to warrant treatment as cognitive agents?
14. How should different temporal rates be governed when fast agents can perform years of subjective work during minutes of human time?
15. Can a system be extremely intelligent but systematically unwise because its objectives, social model, or epistemic incentives are poor?
16. What is the appropriate unit of analysis when cognition spans model, tools, memory, humans, organizations, and environment?
17. Which physical limits matter in practice long before fundamental thermodynamic limits are reached?
18. How can provenance and causal traceability survive across federated agents, model updates, forks, and merges?

19. What forms of intelligence would be invisible to benchmarks built around human language and occupational tasks?
20. Which categories in this report are artifacts of present-day human conceptual habits rather than genuine joints in the space of possible minds?

24. Research Curriculum and Simulation-Lab Exercises

24.1 Eight-stage learning path

Stage 1 — Definitions

Read competing intelligence definitions and write three incompatible operational definitions. Goal: become comfortable with pluralism rather than hunting for one slogan.

Stage 2 — Generalization

Study transfer, few-shot learning, ARC-style reasoning, continual learning, and benchmark limitations. Goal: distinguish knowledge from skill acquisition.

Stage 3 — Agency

Build conceptual diagrams separating model, memory, planning loop, tools, environment, and permissions.

Stage 4 — Architecture

Compare monolithic, modular, neuro-symbolic, multi-agent, embodied, and evolutionary systems.

Stage 5 — Consciousness and identity

Study consciousness indicators, self-models, personal identity, fork/merge problems, and moral uncertainty.

Stage 6 — Collective intelligence

Study teams, swarms, institutions, markets, and civilizations as information-processing systems.

Stage 7 — Physical limits

Model latency, energy, bandwidth, compute, memory, and fabrication as constraints on intelligence.

Stage 8 — Worldbuilding synthesis

Create two intelligences with equal “overall impressiveness” but orthogonal Intelligence Architecture Profiles; test whether the setting can explain their differences without using a single IQ-like rank.

24.2 Eight simulation exercises

Exercise 1: Novel Domain Transfer

Research question	Can a broad system learn an unfamiliar rule world from sparse examples?
Simulated environment	Procedurally generated environment with withheld concepts.
Measure	Learning efficiency, calibration, transfer, failure recognition.

Safety boundary	No outside tools until baseline is measured.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 2: Federation vs. Monolith

Research question	When does a multi-agent federation outperform one model?
Simulated environment	Same research problem assigned to monolith and specialist council.
Measure	Accuracy, diversity, communication cost, correlated errors.
Safety boundary	Equal compute budgets where possible.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 3: Fork-and-Merge Identity

Research question	What is lost or gained when one agent branches?
Simulated environment	Three forks receive incompatible experiences and later reconcile.
Measure	Knowledge integration, value drift, conflict rate, identity claims.
Safety boundary	No external real-world actions.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 4: Slow Oracle

Research question	Can long-horizon reasoning remain useful under changing assumptions?
Simulated environment	Simulated century with periodic environmental shocks.
Measure	Plan robustness, update latency, assumption decay.
Safety boundary	Regular reality-check intervals.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 5: Embodied Generalization

Research question	Does abstract competence transfer to physical control?
Simulated environment	Robotic simulator with novel objects and sensor noise.
Measure	Planning success, safety margin, recovery from errors.
Safety boundary	Bounded actuator force and rollback.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 6: Ecological Intelligence

Research question	Can useful group cognition emerge without central control?
Simulated environment	Artificial-life ecology with heterogeneous agents.
Measure	Novelty, stability, cooperation, complexity, resource dynamics.
Safety boundary	Hard resource and containment boundaries.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 7: Adversarial Epistemics

Research question	Can an intelligence preserve truth-tracking under coordinated misinformation?
Simulated environment	Multi-agent information environment with planted false sources.
Measure	Calibration, provenance use, contradiction detection.
Safety boundary	No privileged hidden answer channel for main agent.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

Exercise 8: Civilization as Mind

Research question	At what integration level does a population behave like a coherent problem solver?
Simulated environment	Layered society with agents, institutions, archives, markets, and shared crises.
Measure	Response quality, memory, coordination, dissent, adaptation.
Safety boundary	Compare centralized and federated governance variants.
What it cannot establish	Simulation performance alone cannot prove consciousness, real-world safety, or universal generality.

25. Final Synthesis

1. Is “general intelligence” actually one thing?

Not conclusively. General factors and broad-capability constructs can be useful, but research definitions emphasize different objects: goal achievement, breadth, performance, transfer, adaptation, or skill-acquisition efficiency. For design purposes, treat generality as a family of properties rather than a metaphysical essence.

2. Is intelligence meaningfully representable on a single scale?

Sometimes for a carefully defined population and task battery, but not reliably for the full space of possible artificial minds. A scalar can summarize; it cannot replace the underlying profile.

3. Which dimensions matter most for artificial-mind design?

Breadth, depth, adaptation, reliability, agency, planning horizon, memory, embodiment, distribution, multiplicity, epistemic discipline, social cognition, value architecture, resource budget, and self-modification are particularly high leverage.

4. What becomes conceivable once human cognition is no longer the template?

Forkable and mergeable minds, multi-body agents, simulation-native organisms, protocol minds, heterogeneous temporal systems, swarms, cognitive institutions, ecological intelligences, and civilization-scale architectures.

5. What separates a machine from an artificial organism, institution, civilization, or cognitive ecosystem?

There is no universal bright line. Increasing persistence, self-maintenance, reproduction, adaptation, embodiment, ecological participation, institutional continuity, and population-level organization can shift which explanatory vocabulary is most useful.

6. When does “an AI” stop being a useful unit of analysis?

When capability depends irreducibly on interacting models, tools, memory systems, humans, institutions, infrastructure, and environments, or when identity continuously forks and merges. At that point the unit may need to become an architecture, ecology, or institution.

7. Which current scientific concepts are most useful for fictional intelligence?

Generalization and learning efficiency, multi-agent systems, embodied cognition, collective intelligence, continual learning, neuro-symbolic architectures, artificial life, information thermodynamics, and theory-driven consciousness indicators.

8. Which speculative concepts are plausible enough for simulation?

Federations, swarms, fork/merge workflows, long-horizon agents, simulation-native systems, artificial-life ecologies, infrastructure cognition, and heterogeneous temporal architectures can all be explored as simulations without claiming they exist as mature real-world systems.

9. Which concepts require new theory rather than simply more compute?

Open-ended evolution, robust continual self-modification, reliable machine metacognition, machine consciousness assessment, group identity, mergeable personal identity, civilization-scale cognition, and governance across radically different mind architectures all appear theory-limited as well as resource-limited.

10. What questions do existing terms prevent us from asking?

They can make us ask “What comes after AGI?” when the deeper question is “What kinds of minds become possible when architecture, scale, embodiment, time, identity, and social organization are free variables?”

Closing framework

AGI should be treated as an important location in the map of possible intelligence, not the destination of the map. “Beyond AGI” is therefore not one rung called ASI. It is a branching design space containing different kinds of minds, collectives, institutions, ecologies, and potentially forms for which “an AI” is no longer the right noun.

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Methodological Note

This is an independent synthesis prepared by ChatGPT. It does not reproduce or rank the other AI responses previously collected by the user. Real-world claims are anchored to the sources above, while the Intelligence Architecture Profile, novel categories, archetypes, institutional applications, universe contrasts, and simulation exercises are original synthesis created for this report. The boundary between evidence and extrapolation is intentionally visible.